Analytical Cume
September 15, 2000

There are three basic methods for determining an unknown concentration using a set of standards, calibration curve, standard addition and internal standard. Answer the questions below based on your basic knowledge of each procedure.

I. Calibration Curve

Absorbance at 1250 cm^{-1} for acetone in solutions in water are shown below:

Volume % acetone	Corrected Absorbance (A)
10	0.215
20	0.411
30	0.578
40	0.745
50	0.913

The resulting equation of the calibration curve is given by:

$$A = 0.01730\ (\pm 0.00033)\ \text{Vol\%} + 0.053(\pm 0.0104)$$

(12 pts) a) Find the volume percent of acetone in a solution whose absorbance is 0.611

(12 pts) b) If the uncertainty in the measured absorbance is ±0.0104, what is the uncertainty in the calculated percentage of acetone, rounded off to the correct number of significant figures?

II. Standard Addition

An unknown sample of Ni^{2+} gave a current of 2.36μA in an electrochemical analysis (assume current is linearly proportional to concentration of Ni^{2+}). When 0.5 mL of a solution containing 0.0287 M Ni^{2+} was added to 25.0 mL of unknown, the current increased to 3.79μA.

(12 pts) a) Derive the expression:

$$\frac{[Ni^{2+}]_i}{[S]_f + [Ni^{2+}]_f} = \frac{\mu A_{Ni^{2+}}}{\mu A_{S+Ni^{2+}}}$$

Where $[Ni^{2+}]_i$ is the concentration of the unknown, and $[Ni^{2+}]_f$ is the concentration of the unknown in the sample diluted by the addition of standard and $[S]_f$ is the